

JomExam

ESAT Chemistry — Sample Paper

Six questions in the style of ESAT Chemistry, including a graph, a paired-options question and a statement-set question. Allow 9 minutes — the real paper's 90-second pace. No calculator. One option is correct in each question, and there is no negative marking.

Worked solutions with the trap named follow the questions. The full experience — 27 questions, 40 minutes, and a report naming the skill behind every wrong answer — is free at jomexam.com.

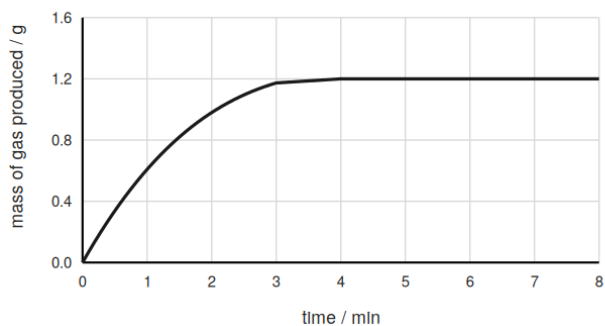
Question 1. Element X has the electron configuration 2.8.6. What is the formula of the compound it forms with sodium?

- A NaX
- B NaX₂
- C Na₂X
- D Na₆X
- E Na₂X₃

Question 2. What is the mass of 0.50 mol of sodium carbonate, Na₂CO₃? (A_r: Na 23, C 12, O 16)

- A 45 g
- B 53 g
- C 106 g
- D 62 g
- E 23 g

Question 3. An excess of marble chips is added to dilute hydrochloric acid, and the mass of gas produced is recorded against time, as shown. At which time is the reaction just complete?



- A 1 min
- B 2 min
- C 4 min
- D 6 min
- E 8 min

Question 4. An excess of zinc is added to 50 cm³ of hydrochloric acid of concentration 2.0 mol dm⁻³: Zn + 2HCl → ZnCl₂ + H₂. What volume of hydrogen is produced at room temperature and pressure? (1 mol of gas occupies 24 dm³ at RTP)

- A 1.2 dm³
- B 2.4 dm³
- C 0.6 dm³
- D 4.8 dm³
- E 0.05 dm³

Question 5. Substance X has a high melting point. It does not conduct electricity when solid, but conducts well when molten. Which row gives the structure of X, and the correct reason?

	Structure of X	Reason
A	simple molecular	weak forces between molecules break on melting
B	giant ionic	the ions are free to move in the solid
C	giant covalent	strong bonds throughout the whole structure
D	metallic	delocalised electrons carry charge in both states
E	giant ionic	ions are held in the solid lattice, and are free to move only when molten
F	simple molecular	covalent bonds break when the solid melts
G	giant covalent	electrons are freed from the bonds on melting
H	metallic	the electrons are free only in the liquid state

Question 6. Consider the following statements about the Group 1 metals:

- Potassium reacts more vigorously with water than sodium does.
- The Group 1 metals form ions with a 2+ charge.
- Reactivity increases down the group because the outer electron is further from the nucleus and more easily lost.

Which statements are correct?

- A none of them
- B 1 only
- C 2 only
- D 3 only
- E 1 and 2 only
- F 1 and 3 only
- G 2 and 3 only
- H 1, 2 and 3

Worked solutions

Question 1 — answer C.

1. 2.8.6 is six outer electrons: X gains two, forming X^{2-} .
2. Sodium forms Na^+ : two sodiums balance one X.

The trap: The trap is NaX — pairing one metal with one non-metal by habit. The formula is decided by the charges, and the charges are decided by the configuration; nothing else votes.

Question 2 — answer B.

1. M_r of $Na_2CO_3 = 46 + 12 + 48 = 106$.
2. Mass = $0.50 \times 106 = 53$ g.

The trap: Two traps: 106 (the molar mass reported, the 0.50 unused) and 45 (one sodium counted). The subscript 2 and the multiplication both have to survive.

Question 3 — answer C.

1. The reaction is complete when no more gas forms — where the curve first becomes flat.
2. The curve reaches its plateau of 1.2 g at exactly 4 minutes.

The trap: The trap is 8 minutes — the end of the GRAPH read as the end of the REACTION. The graph keeps being drawn after the chemistry has stopped; the answer is where the line goes flat, not where it runs out.

Question 4 — answer A.

1. Moles of $HCl = 0.050 \times 2.0 = 0.10$ mol.
2. The 2 : 1 ratio gives 0.050 mol of H_2 .
3. Volume = $0.050 \times 24 = 1.2$ dm³.

The trap: The trap is 2.4 dm³ — the ratio skipped, one hydrogen per acid. The 2 in the equation is a divide-by-two in the arithmetic, and it is the only step the question is really about.

Question 5 — answer E.

1. Conducts molten but not solid: the charge carriers exist in both states but move in only one — ions in a lattice.
2. The reason must say the ions are FREED by melting, not created by it.

The trap: Row B is the harvest: right structure, wrong reason — ions are NOT free in the solid, which is precisely why it does not conduct. In paired questions the reason column fails more candidates than the structure column.

Question 6 — answer F.

1. Statement 1: reactivity increases down Group 1 — true.
2. Statement 2: Group 1 forms 1+ ions — false.
3. Statement 3: the standard mechanistic explanation — true. So 1 and 3 only.

The trap: Statement 2 is the planted near-miss: one wrong digit in a familiar fact. Statement sets reward reading each claim as a checkable sentence rather than as scenery.